

Multiple Choice (10 marks)

1. In a _____ attack, the extra data that holds some specific instructions in the memory for actions is projected by a cyber-criminal or penetration tester to crack the system.
 - (a) Phishing
 - (b) MiTM
 - (c) Buffer-overflow
 - (d) Clickjacking
2. Why would a ping sweep be used?
 - (a) To identify live systems
 - (b) To locate live systems
 - (c) To identify open ports
 - (d) To locate firewalls
3. What are the port states determined by Nmap?
 - (a) Active, inactive, standby
 - (b) Open, half-open, closed
 - (c) Open, filtered, unfiltered
 - (d) Active, closed, unused
4. Applications developed by programming languages like _____ and _____ have this common buffer-overflow error.
 - (a) C, Ruby
 - (b) Python, Ruby
 - (c) C, C++
 - (d) Tcl, C#
5. Can a stream cipher have perfect secrecy?
 - (a) Yes, if the PRG is really "secure"
 - (b) No, there are no ciphers with perfect secrecy
 - (c) Yes, every cipher has perfect secrecy
 - (d) No, since the key is shorter than the message

True / False (10 marks)

Answer True or False

6. True or False: A stack is a sequential segment of memory that stores data in a manner similar to a buffer, allowing for random access.
7. True or False: The four primary security principles related to messages are confidentiality, access control, integrity, and non-repudiation.
8. True or False: Wireless access refers to the methods used to connect devices to a wired network for data transmission.
9. True or False: Authorization is primarily focused on determining the identity of the user and whether they are an attacker.
10. True or False: Applications developed in C and C++ are prone to buffer overflows due to the lack of string boundary checks in predefined functions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the occurrence and position of the substrings within a string.
12. Write a function to find the directrix of a parabola.

End of Paper